SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0510**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

I P. Akhila/ 192472320 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Akhila

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

(a) **Clearly interpret the scenario and explain the cause of low contrast.**
A grayscale image appears dull because the pixel intensity values of the object and the background are very close to each other. Since there is very little difference in brightness, the object cannot be distinguished clearly from the background. Low contrast usually occurs due to poor lighting conditions, improper camera settings, or limited intensity variation in the image.

(b) **Identify all key issues affecting image visibility.**

- Object and background have similar intensity values.
- Important image details are difficult to identify.
- Edges are not clearly visible.
- Image quality is poor, making further processing difficult.

(c) **Apply an appropriate enhancement technique to address the problem.**
Histogram Equalization
Histogram Equalization redistributes the pixel intensity values over the entire intensity range. This increases the contrast between dark and bright regions, making objects easier to see.

(d) **Justify your choice with logical reasoning.**
Histogram Equalization is suitable because it automatically improves image contrast without requiring manual adjustment. It enhances image details and makes feature extraction and segmentation more accurate.

(e) **Present your explanation in a clear and structured manner.**
Therefore, Histogram Equalization is the best technique for improving a low-contrast image because it enhances visibility and increases the difference between the object and the background.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

(a) **Interpret the scenario and explain the impact of uneven illumination.**
The image contains bright and dark regions because the lighting is not uniform during image capture. This results in some parts appearing overexposed while others appear too dark.

(b) **Identify key factors causing intensity variation.**
Non-uniform lighting

Shadows
Unequal illumination across the scene
Improper camera exposure

(c) **Apply a suitable enhancement method to normalize illumination.**
Adaptive Histogram Equalization (CLAHE)
CLAHE improves contrast locally by processing small regions of the image instead of the whole image.

(d) **Justify your selected approach with reasoning.**
CLAHE enhances dark regions while preventing excessive brightness in already bright regions. It produces a balanced image even under uneven lighting conditions.

(e) **Provide a clear and well-structured explanation.**
CLAHE effectively normalizes illumination and improves the visibility of image details.

3. Salt-and-Pepper Noise
A scanned document contains salt-and-pepper noise affecting readability.

(a) **Explain the scenario and characteristics of this noise.**
Salt-and-pepper noise appears as random black and white pixels scattered throughout an image. It usually occurs because of sensor errors, faulty image transmission, or scanning problems. This noise reduces the readability of scanned documents.

(b) **Identify key issues impacting image quality.**
- Image becomes noisy.
- Text and edges become unclear.
- Small details are lost.
- Image quality decreases.

(c) **Apply an appropriate filtering technique.**
Median Filter
A Median Filter replaces each pixel with the median value of its neighboring pixels.

(d) **Justify why this method is more suitable than alternatives.**
The Median Filter removes salt-and-pepper noise effectively while preserving image edges. Unlike the averaging filter, it does not blur important details.

(e) **Present your response clearly and logically.**
The Median Filter is the most suitable method because it removes impulse noise and preserves image quality.

4. Blurred Edges
An image processed for noise removal appears overly smooth, causing edges to blur.

(a) **Interpret the scenario and explain why edges are blurred.**
The image has undergone excessive smoothing during noise removal. As a result, sharp edges have become blurred, making object boundaries difficult to identify.
(b) **Identify key issues related to smoothing and edge loss.**
- Edge information is lost.
- Image appears overly smooth.
- Object boundaries become unclear.
- Feature detection accuracy decreases.

(c) **Apply an appropriate technique to restore edges.**
Laplacian Sharpening Filter
The Laplacian filter highlights regions where intensity changes rapidly, making edges sharper.

(d) **Justify your decision with clear reasoning.**
Sharpening restores lost edge information and improves image clarity after smoothing. It makes object boundaries more distinct.

(e) **Structure your explanation clearly.**
Applying the Laplacian filter improves edge visibility and restores important image details.

5. High-Frequency Noise
A satellite image contains fine-grained noise affecting feature visibility.
(a) **Interpret the scenario and describe high-frequency noise.**
A satellite image contains fine-grained noise that appears as rapid intensity changes across the image. This type of noise makes it difficult to identify important features such as roads, buildings, or vegetation.
(b) **Identify the key issues affecting image clarity.**
- Fine-grained noise reduces image clarity.
- Important features become difficult to identify.
- Image interpretation becomes less accurate.
- Feature extraction performance decreases.

(c) **Apply an appropriate frequency domain filtering method.**
Low-Pass Filter (LPF)
A Low-Pass Filter removes high-frequency components while preserving the important low-frequency information.

(d) **Justify the choice of filter with reasoning.**
High-frequency noise mainly exists in the higher frequency components of an image. A Low-Pass Filter suppresses these unwanted frequencies, resulting in a smoother and clearer image.

(e) **Present your answer in a clear and organized manner.**
The Low-Pass Filter is the most suitable choice because it effectively removes high-frequency noise while preserving the important information in the satellite image.